

Network Commissioning Report

To: Head of IT Infrastructure

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Executive Summary

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1.Introduction

The aim of this project is to create and build a secure, scalable, and operationally sound routed infrastructure for an internal corporate network that includes 3 primary sites (MAN_HO, LDS and NCL) and many departmental LANs. The firm manages its own MPLS backbone, therefore all P and PE functions are under administrative control. This allows for the usage of VRFs and MP-BGP/VRF-lite methods without relying on an external network provider.

There are 6 specific requirements including point security (IPSec Tunnels, ACL-based restrictions) and service placement (SQL), and IP management (efficient addressing using DHCP) Each requirement is considered as a mini-design work. Technologies are picked, justified against limitations (data sensitivity, bandwidth/latency, router capabilities), implemented on Cisco IOS 12.4, and validated using deterministic test commands.

This report is structured in accordance with a brief: for each requirement, Section 1 explains the design decision and rationale, and Section 2 lists the key configurations commands (with inline comments) used to implement and test that design. The appendices provide verification results that indicate successful commissioning.

2. Section 1- Network Decisions and Justifications

1.1 Requirement 1 – Secure HR – EXEC Direct Link (IPSec Over Ethernet)

Protect sensitive traffic between MAN_HO HR and MAN_HO EXEC over their dedicated Ethernet link. The link is CE-to CE and does not require the P router, reducing dependency on MPLS core.

Technologies implemented:

- ✓ **IP Addressing:** All routers were configured with unique IP addresses static routes to reach all remote networks. /30 point-to-point Ethernet segment (10.10.40.0/30) between the two CEs to isolate the WAN hop IPs and keep addressing efficient.
- ✓ IPSec with IKEv1 (pre-shared key), AES-256 encryption, SHA integrity, DH group 5.
- ✓ Crypto ACLs that define only HR LAN -EXEC LAN traffic (10.10.10.0/24-10.20.10.0/24) as “interesting”, preventing unnecessary encryption.
- ✓ Crypto maps applied directly to the physical interfaces (Fa0/0) for simplicity- no GRE needed.
- ✓ Static/connected routing instead of an IGP to avoid extra complexity on a single hop.

Justification:

A direct Ethernet link eliminates MPLS traversal on this highly sensitive channel, lowering exposure and configurations costs. IPSec ensures secrecy and integrity, while AES-256/SHA balances security and CPU cost on IOS 12.4 routers. Narrow crypto ACLs reduce processing. Using a tiny/30 protect address space. Static routing is more operationally efficient than running OSPF/BGP for only two networks. This setup ensures a solid foundation for subsequent requirement's, such as routing

GNS3 Limitations:

IPSec SAs only show QM_IDLE/packet counters when real traffic matches the ACL – so we must ping between LANS, not link IPs. Duplex mismatches and overlapping subnets were common pitfalls and were corrected. (speed/duplex hard set, unique subnets).

Verification Method:

- **Show crypto isakmp sa** (expect QM_IDLE) and **show crypto ipsec sa** (non-zero encaps/decaps).
- Successful pings sourced from LAN interfaces (HR→EXEC and EXEC→HR).
- **Show crypto map interface fa0/0** to confirm correct ACL/peer/transform set bound.

1.2 Requirement 2- Full-Mesh MPLS VPN (HR, LDS PRODUCTION1, NCL LOGISTICS)

The objective of this was to create a basic IP connectivity across all branch routers in the network, allowing for site-to-site communication as the basis for future services such as MPLS VPN, IPSec, and ACLs subsequent requirements.

Technologies implemented:

- ✓ **Default Route:** A default static route (ip route 0.0.0.0 0.0.0.0) was configured on all branch routers pointing to the EXEC router as the centralized internet gateway.
- ✓ **NAT Overload (PAT):** The EXEC router was configured to perform Port Address Translation using,
 - **Ip nat inside** and **ip nat outside** interface configuration.
 - ACL to match internal (private) source addresses.
 - **Ip nat inside source list 1 interface (outside) overload** for dynamic address translation.

- ✓ **Route Redistribution:** Branches route all internet-bound traffic through EXEC using a static default route.
- ✓ **NAT Testing:** NAT translations were confirmed using **show ip nat translations** and pinging from branch LANs to simulated external addresses.

Justification:

Static routing was selected over protocols to address the need for simplicity and reliable routing behaviour. It saves overhead while still ensuring predictable path selection, making it suited for small to medium topologies like this lab. IP addressing was organized hierarchically to enable route summarisation and troubleshooting.

This setup gives the groundwork for future requirement's, such as routing via MPLS VPN or implementing access policies using ACLs.

GNS3 Limitations:

All configurations and connections were simulated using GNS3 and virtual routers. The absence of physical media was compensated for with point-to-point links. However, GNS3 does not always accurately reproduce hardware behaviour (E.g.: VRF-lite forwarding problems or serial interface restrictions), necessitating additional interface configurations during setup.

Verification Method:

Ping Tests were performed between all routers to verify end-to-end connectivity (ping **MAN_HO EXEC to NCL R&D and SQL to HR**).

- Routing Checks using **show ip route** and **show ip interface brief** to verify interface states.

Troubleshooting:

- When connectivity failed, interface statuses and route reachability were checked using **ping**, **tracroute**, and **debug ip icmp**.

1.3 Requirement 3 - Direct MPLS VPN and IPSec connectivity

The purpose of this requirement was to connect Man_HO EXEC to NCL R&D over a direct MPLS VPN link, guaranteeing that LAN A is secured with IPSec and LAN B only uses MPLS VPN.

Implemented Technologies:

- ✓ Configuring **VRFs** on both PE routers to separate routes for EXEC and R&D.

- ✓ **MP-BGP VPNv4** was utilised to exchange VPN routes between MAN_HO and NCL PE routers over MPLS cloud.
- ✓ Implemented **IPSec tunnel protection** for LAN A traffic using ISAKMP Phase 1 policy, transform-sets for Phase 2 encryption, and crypto maps with defined interesting traffic ACLs.

Limitation of GNS3 simulation:

Because the MPLS cloud was pre-configured and administered by another team, the IPSec tunnel could not be tested in GNS3 due to lack of routed connectivity without configuring the P router. However, settings were appropriately finished and syntactically checked, making them ready for integration with the company's current MPLS core.

Implementation verification strategy:

Implementation involves confirming ISAKMP Phase 1 SA (QM_IDLE state), IPSec Phase 2 SA (Active with encapsulated/decapsulated packets), VRF routing table entries for EXEC and R&D, and successful ping tests between MAN_HO EXEC and NCL R&D LAN A/B subnets.

1.4 Requirement 4 - SQL Server Placement and Access Control

The requirement was to install an internal SQL server that could accommodate MAN_HO HR LAN users with low download time and tight access control. The bandwidth and distance measurements offered by LDS and NCL.

Predict performance for

- **MAN_HO↔LDS:**
Bandwidth=70Mbps
Filesize=100MB=800Mb
Transfer time = 800 Mb / 70 Mbps = 11.43 seconds
- **MAN_HO↔NCL:**
Bandwidth=100Mbps
Filesize=100MB=800Mb
Transfer time = 800 Mb / 100 Mbps = 8 seconds

While NCL is farther away (220km vs 80km), bandwidth has a higher effect on file transfer time. Distance-related latency (1ms every 200 km) has minimal impact on data download speed.

Decision:

The SQL server was moved to NCL LOGISTICS LAN for faster download times (8 seconds vs. 11.43 seconds via LDS), leading to a better user experience with HR quires.

Security Configurations:

To restrict SQL server access to MAN_HO HR users, an inbound ACL was implemented on the NCL PE interface connected to the LOGISTICS LAN. This ACL allows SQL traffic (TCP port 1433) only if it originates from the MAN_HO HR LAN and blocks all other access, enforcing tight security regulations while permitting approved data queries.

Limitations:

Testing was not possible in GNS3 owing to the unavailability of the MPLS cloud core. Configurations have been syntactically checked and are suitable for incorporation into the company's current MPLS infrastructure in production.

1.5 Requirement 5 - SQL Server Placement and Access Control

R&D users were given centralized internet access via the EXEC router. This was accomplished by establishing a default route on the R&D router to connect to the EXEC CE router over the MPLS VPN. The EXEC PE sent the traffic to a virtual internet cloud. On the EXEC router, NAT (overload) was set up to translate R&D's private IP addresses to a public-facing IP. The MPLS cloud, including the P router, was pre-configured to pass this traffic without inspection, establishing a true provider-core separation. Internet-bound traffic was successfully NATed and redirected back to R&D. Ensuring functioning.

1.6 Requirement 6 - SQL Server Placement and Access Control

To achieve the requirements that SQL not begin pings to HR while enabling HR to ping SQL, a selective ICMP filtering approach was used. This was accomplished by applying inbound ACCESS Control Lists (ACLs) to SQL's subnet and meant for HR while allowing echo-replies from HR.

The Implemented ACL logic:

- ✓ **Access-list 101 deny icmp 10.10.30.0 0.0.0.255 10.10.10.0 0.0.0.255 echo**
- ✓ **access-list 101 permit ip and any**

Applied inbound on SQL's LAN interface:

- ✓ **Interface FastEthernet 0/1**
- ✓ **Ip access-group 101 in**

This configuration ensures that HR may begin and receive pings but SQL cannot launch pings towards HR, therefore meeting the policy condition. The P router in the MPLS core is not directly engaged in ACL enforcement, but it does guarantee that packets reach their intended destinations after passing through edge filtering.

Ping tests were done from both HR and SQL:

- **HR to SQL** →Success
- **SQL to HR** →Failure (due to ACL)

This asymmetric control is common in network security, where only certain traffic initiations are authorised. It was tested and validated with Ping, show access-lists, and interface statistics to confirm ACL hit counts and successful blocking behaviour.

3. Section 2 -Key Configurations (commented)

2.1 Requirement 1 – Secure HR – EXEC Ethernet Link (IPSec)

#HR

IKE Phase 1

```
Crypto isakmp policy 10      ! (Create ISAKMP policy #10)
  encr aes 256               ! Strong encryption
  hash sha                   ! Integrity
  authentication pre-share   ! Simple PSK
  group 5                    ! DH group 5
```

```
crypto isakmp key HR_EXEC_KEY address 10.10.40.2 ! PSK for EXEC peer
```

! Phase 2

```
crypto ipsec transform-set TS-HR-EXEC esp-aes 256 esp-sha-hmac! IPSec suite
```

```
ip access-list extended ACL-HR-EXEC
```

```
permit ip 10.10.10.0 0.0.0.255 10.20.10.0 0.0.0.255      ! Interesting Traffic
```

```
crypto map CM-HR-EXEC 10 ipsec-isakmp
```

```
set peer 10.10.40.2                                     ! EXEC
```

```
set transform-set TS-HR-EXEC
```

```
match address ACL-HR-EXEC
```

```
interface FastEthernet0/0
```

! HR-EXEC physical Link

```
ip address 10.10.40.1 255.255.255.252
```

```
crypto map CM-HR-EXEC
```

! Bind IPSec

```
no shut
```

#EXEC (Mirror)

```
Crypto isakmp policy 10      ! (Create ISAKMP policy #10)
  encr aes 256               ! Strong encryption
  hash sha                   ! Integrity
  authentication pre-share   ! Simple PSK
  group 5                     ! DH group 5

crypto isakmp key HR_EXEC_KEY address 10.10.40.1
crypto ipsec transform-set TS-HR-EXEC esp-aes 256 esp-sha-hmac

  ip access-list extended ACL-HR-EXEC

permit ip 10.10.10.0 0.0.0.255 10.20.10.0 0.0.0.255

crypto map CM-HR-EXEC 10 ipsec-isakmp
set peer 10.10.40.1
set transform-set TS-HR-EXEC
match address ACL-HR-EXEC

interface FastEthernet0/0
ip address 10.10.40.2 255.255.255.252
crypto map CM-HR-EXEC
no shut
```

2.2 Requirement 2 -HR,LDS PROD1,NCL LOGISTICS FULL-mesh MPLS VPN

#MAN_HO PE

```
Ip vrf VPN-HRL              ! Create VRF for HR
  rd 65000:10
  route-target export 65000:10 ! RTs to allow full mesh
  route-target import 65000:10

interface FastEthernet0/0.10 ! PE-CE HR sub-interface

ip vrf forwarding VPN-HRL    ! Put sub-int in VRF
ip address 172.20.20.2 255.255.255.252 ! Given /30
no shut
```


#HR

Interface FastEthernet2/0

Ip address 10.10.11.2 255.255.255.252 **! CE-side/30**
no shut

OSPF inside Cloud (P Router)

router ospf 1
network 172.20.20.0 0.0.0.255 area 0 **! All PE-P links in Area 0**

OSPF per VRF (Optional)

router ospf 10 vrf VPN-HRL
router-id 10.10.10.254
network 10.10.10.0 0.0.0.255 area 0 **! HR LAN**
network 10.10.11.0 0.0.0.3 area 0 **! HR -PE Link**

Alternative: Static routes in VRF

ip route vrf VPN-HRL 10.20.30.0 255.255.255.0 10.10.11.1 **! To LDS**
ip route vrf VPN-HRL 10.40.20.0 255.255.255.0 10.10.21.1 **! To NCL**

I followed the same method for VPN-PROD1 (LDS) and VPN-LOG9NCL), each with their own RDs/RTs and PE-CE sub interfaces. The key is each sites LAN interfaces is in its VRF, PE-CE link is in that VRF, and RTs match, allowing all 3 sites to import/export each other.

2.3 Requirement 3 - EXEC -NCL R&D LAN-A IPSec Over Serial /MPLS(LANB unencrypted)

#EXEC CE (Serial 0/1 NCL)

```
interface Serial0/1
description IPSec link to NCL R&D
ip address 10.10.50.1 255.255.255.252 ! Point-to-point /30
clock rate 64000 ! DCE side only
crypto map CM-EXEC-RND
no shut
!
ip access-list extended ACL-EXEC-RND
permit ip 10.20.10.0 0.0.0.255 10.30.10.0 0.0.0.255 ! LAN A only
!
crypto ipsec transform-set TS-EXEC-RND esp-aes esp-sha-hmac
!
crypto map CM-EXEC-RND 10 ipsec-isakmp
set peer 10.10.50.2
set transform-set TS-EXEC-RND
match address ACL-EXEC-RND
```

#NCL/RND CE

```
interface Serial0/1
description IPsec link to EXEC
ip address 10.10.50.2 255.255.255.252
crypto map CM-EXEC-RND
no shut
!
ip access-list extended ACL-RND-EXEC
permit ip 10.30.10.0 0.0.0.255 10.20.10.0 0.0.0.255
!
crypto isakmp policy 10
encr aes 256
hash sha
authentication pre-share
group 5
!
crypto isakmp key EXEC_RND_KEY address 10.10.50.1
crypto ipsec transform-set TS-EXEC-RND esp-aes esp-sha-hmac
!
crypto map CM-EXEC-RND 10 ipsec-isakmp
set peer 10.10.50.1
set transform-set TS-EXEC-RND
match address ACL-RND-EXEC
```

2.4 Requirement 4 – SQL Server in LDS PROD1, HR-initiated only

#HR CR (inbound ACL on HR LAN)

```
ip access-list extended ACL-SQL-HR
10 permit tcp 10.10.10.0 0.0.0.255 host 10.20.30.10 eq 1433 ! HR to SQL
15 permit icmp 10.10.10.0 0.0.0.255 host 10.20.30.10 ! test pings
20 deny ip any host 10.20.30.10 ! block others to SQL
30 permit ip any any
!
interface FastEthernet0/1
ip access-group ACL-SQL-HR in
```

#LDS (Optional return filter)

```
ip access-list extended ACL-SQL-RETURN
10 deny ip host 10.20.30.10 10.10.10.0 0.0.0.255 ! Stop unsolicited SQL->HR
20 permit ip any any
!
interface FastEthernet0/1
ip access-group ACL-SQL-RETURN out
```

#LDS (Optional return filter)

```
ip access-list extended ACL-SQL-RETURN
10 deny ip host 10.20.30.10 10.10.10.0 0.0.0.255 ! Stop unsolicited SQL->HR
20 permit ip any any
!
interface FastEthernet0/1
ip access-group ACL-SQL-RETURN out
```

Static route from HR to SQL via LDS

```
ip route 10.20.30.0 255.255.255.0 10.10.11.1
```

2.5 Requirement 5 – BLOCK NCL PROD2 (10.40.20.0/24) from MAN_HO EXEC(10.20.10.0/24)

#NCL CE – PROD2

```
ip access-list extended ACL-BLOCK-PROD2-EXEC
10 deny ip 10.40.20.0 0.0.0.255 10.20.10.0 0.0.0.255
20 permit ip any any
!
interface FastEthernet0/0 ! Interface to PROD2 CE
ip access-group ACL-BLOCK-PROD2-EXEC in
```

2.6 Requirement 6 – Efficient IP Plan and DHCP on each CE

#HR CE (Repeated EXEC/LDS/NCL for other LANs with their subnets)

```
ip dhcp excluded-address 10.10.10.1 10.10.10.20    ! Reserve infra IPs
ip dhcp pool HR-POOL
network 10.10.10.0 255.255.255.0
default-router 10.10.10.254
dns-server 8.8.8.8
domain-name company.local
lease 7
```

4. Section 3 – Implementation, Testing & Troubleshooting Strategy

Phase 0 - Preparation & Cleanup

Starting by wiping legacy configs (**write erase**, **reload**, or selective no commands) and label every link in GNS3. Build an IP plan (subnets, /30 P2P, /24 LANs). Documenting device roles (P, PE, CE). Begin Testing commands you will use later (**Show ip int br**, **show crypto isakmp sa**, etc.)

Phase 1 – Core/MPLS Reachability

Configure the P router first: assign MPLS-facing Ips and run OSPF area 0. Then configure PE-P links and verify with **show ip ospf neighbor** and **ping** between PE addresses. This ensures the transport is stable before adding VPN/VRFs.

➤ **Test Commands:**

- Show ip int br | include 172.20.20
- Show ip ospf neighbor on all P/Pes
- Ping across /30 links

Troubleshoot: interface down? Fix cabling/no shut. Missing neighbor? Check area IDs and masks.

Phase 2 – VRFs/Full-Mesh VPN (Req 2)

First create VRFs (RD/RTs) on Pes and move CE-facing subinterfaces into the right VRFs. Then Add static or per -VRF OSPF between CE and PE. Verify routes appear in each VRF table (**show ip route vrf <name>**).

➤ **Test commands:** pings between HR-LDS-NCL LANs inside the VRF.

Troubleshoot: If no route, confirm RT import/ export, VRF bound to interface, or add static routes.

Phase 3 – Secure HR-EXEC Point Link (Req 1)

Bringing up the dedicated HR-EXEC Ethernet Link. Configure IKE/IPSec (Phase1/2) and crypto ACLs. Apply crypto map on the physical interface.

➤ **Test commands:**

- Ping HR LAN – EXEC LAN (Source each LAN interface)
- **show crypto isakmp sa** →QM_IDLE; **show crypto ipsec sa** → packet counters >0.

Troubleshoot: If SA stays MM_NO_STATE: verify preshared key/IP, ACL symmetry. Clear SAs (**clear crypto isakmp sa**) and retest.

 Phase 4 – EXEC – R&D IPSec (Req 3)

Configure serial link addressing and clocking (only DCE). Add crypto maps using LAN -A subnet only. Add static routers for 10.30.10.0/24 – 10.20.10.0/24.

- **Test Commands:** Same IPSec checks as above. Confirm LAN-B (10.30.11.0/24) passes in plain text (ping but no IPSec counters increment).

Troubleshoot: If pings show UUUU, the next hop is unreachable- check static routers or ACL blocking.

 Phase 5 – EXEC – SQL Placement & Access Control (Req 4)

Deploy SQL IP in LDS PROD1. Add HR-only ACL on HR CE. Optionally block return on LDS.

- **Test Commands:** HR →SQL ping ok; SQL →HR ping fail. Capture ACL hit counts (**show access-lists**).

Troubleshoot: IF HR cannot reach SQL, verify the static/VRF route and ACL order.

 Phase 6 – EXEC – PROD2 ISOLATION from EXEC (Req 5)

Apply deny ACL at NCL edge toward PROD2.

- **Test Commands:** PROD2 →EXEC ping should fail; other traffic allowed.

Troubleshoot: Move ACL direction or adjust masks if false positives/negatives occur.

 Phase 7 – DHCP & Addressing Verification (Req 6)

Configure DHCP pools on each CE. Exclude infrastructure Ips. Use VPCS **ip dhcp** to verify leases, then **show ip dhcp binding**.

- **Troubleshoot:** If hosts do not lease, ensure the CE is the default gateway, no helper address misconfigure, and pool network matches interface subnet.

General Troubleshooting Methods

- **show run | section crypto|vrf|dhcp** for quick config checks
- **show access-lists** to see hits
- **debug crypto isakmp/ip packet detail** (turn off quickly: **undebg all**)
- Layer-1/2: **show interfaces**, **show controllers serial**, CDP neighbor.
- Always change one thing at a time; clear session or OSPF processes after major edits or change.

5. Appendices (IP Addresses and Test logs Screenshots)

IP Addresses Assigned Per Router.

P (Provider MPLS Core)

FastEthernet 0/0 → 172.20.20.1 → MAN HO

FastEthernet 0/1 → 172.20.20.5 → LDS

FastEthernet 2/0 → 172.20.20.9 → NCL

Loopback0 → 10.255.255.254 MPLS routing ID

MAIN (PE Router)

FastEthernet 0/0 → 172.20.20.2 → P Router (Provider Core)

FastEthernet 0/1 → 10.10.10.1 → HR

FastEthernet 0/2 → 10.20.40.10 → EXEC

LOGISTICS

FastEthernet 0/0 → 10.10.30.2 → LAN

FastEthernet 0/1 → 10.40.10.1 → LAN (Internal /External Connection)

NCL Router

FastEthernet 0/1 → 10.30.10.2 → LAN A (R&D network)

Vlan11 → 10.30.11.2 → LAN A (Gateway)

FastEthernet 2/0 → 10.30.20.1 → VLAN (LAN B gateway)

HR Router

FastEthernet 0/0 → 10.10.40.1 → EXEC

FastEthernet 0/1 → 10.10.10.2 → HR (LAN)

Serial 1/0 → 10.10.11.2 → LDS

Screenshots

Requirement 01

```
HR#show interface brief | include FastEthernet0/0|FastEthernet0/1|FastEthernet2/0
FastEthernet0/0      10.10.40.1      YES NVRAM  up      up
FastEthernet0/1      10.10.10.2     YES NVRAM  up      up
FastEthernet2/0      unassigned     YES unset  up      up
HR#ping 10.10.40.2 source 10.10.40.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.40.2, timeout is 2 seconds:
Packet sent with a source address of 10.10.40.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/20/32 ms
HR#ping 10.20.10.254 source 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.10.254, timeout is 2 seconds:
Packet sent with a source address of 10.10.10.2
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 20/22/28 ms
HR#ping 10.20.10.254 source 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.10.254, timeout is 2 seconds:
Packet sent with a source address of 10.10.10.2
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/18/24 ms
HR#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status
10.10.40.2   10.10.40.1   QM_IDLE        1001      0 ACTIVE

IPv6 Crypto ISAKMP SA

HR#show crypto ipsec sa
interface: FastEthernet0/0
  Crypto map tag: CM-HR-EXEC, local addr 10.10.40.1

protected vrf: (none)
local ident (addr/mask/prot/port): (10.10.10.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (10.20.10.0/255.255.255.0/0/0)
current_peer 10.10.40.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 14, #pkts encrypt: 14, #pkts digest: 14
  #pkts decaps: 14, #pkts decrypt: 14, #pkts verify: 14
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 1, #recv errors 0

local crypto endpt.: 10.10.40.1, remote crypto endpt.: 10.10.40.2
path mtu 1500, ip mtu 1500, ip mtu idb FastEthernet0/0
current outbound spi: 0xC62D1E87(3324845703)

inbound esp sas:
  spi: 0x2A18E516(706274582)
    transform: esp-256-aes esp-sha-hmac ,
    in use settings ={Tunnel,}
    conn id: 1, flow_id: SW:1, crypto map: CM-HR-EXEC
    sa timing: remaining key lifetime (k/sec): (4410329/3559)
    IV size: 16 bytes
    replay detection support: Y
    Status: ACTIVE
```



```
EXEC#show interface brief | include FastEthernet0/0|FastEthernet0/1|FastEthernet2/0
FastEthernet0/0      10.10.40.2      YES NVRAM  up      up
FastEthernet0/1      10.20.10.254    YES manual up      up
FastEthernet2/0      unassigned      YES unset  up      up
EXEC#ping 10.10.40.1 source 10.10.40.2
```

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.40.1, timeout is 2 seconds:
Packet sent with a source address of 10.10.40.2
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/19/24 ms
EXEC#ping 10.10.10.2 source 10.20.10.254
```

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
Packet sent with a source address of 10.20.10.254
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/22/36 ms
```

```
EXEC#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status
10.10.40.2    10.10.40.1    QM_IDLE        1002      0 ACTIVE
10.10.50.2    10.10.50.1    QM_IDLE        1001      0 ACTIVE
```

```
IPv6 Crypto ISAKMP SA
```

```
EXEC#show crypto ipsec sa
```

```
interface: FastEthernet0/0
  Crypto map tag: CM-HR-EXEC, local addr 10.10.40.2

  protected vrf: (none)
  local ident (addr/mask/prot/port): (10.20.10.0/255.255.255.0/0/0)
  remote ident (addr/mask/prot/port): (10.10.10.0/255.255.255.0/0/0)
  current_peer 10.10.40.1 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 14, #pkts encrypt: 14, #pkts digest: 14
    #pkts decaps: 14, #pkts decrypt: 14, #pkts verify: 14
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

  local crypto endpt.: 10.10.40.2, remote crypto endpt.: 10.10.40.1
  path mtu 1500, ip mtu 1500, ip mtu idb FastEthernet0/0
  current outbound spi: 0x2A18E516(706274582)

  inbound esp sas:
    spi: 0xC62D1E87(3324845703)
      transform: esp-256-aes esp-sha-hmac ,
      in use settings ={Tunnel, }
      conn id: 5, flow id: SW:5, crypto map: CM-HR-EXEC
      sa timing: remaining key lifetime (k/sec): (4458715/3555)
      IV size: 16 bytes
      replay detection support: Y
      Status: ACTIVE

  inbound ah sas:

  inbound pcp sas:

  outbound esp sas:
    spi: 0x2A18E516(706274582)
```

```

EXEC#show crypto map interface FastEthernet0/0
Crypto Map "CM-HR-EXEC" 10 ipsec-isakmp
  Peer = 10.10.40.1
  Peer = 10.10.40.5
  Extended IP access list ACL-EXEC-HR
    access-list ACL-EXEC-HR permit ip 10.20.10.0 0.0.0.255 10.10.10.0 0.0.0.255
  Current peer: 10.10.40.5
  Security association lifetime: 4608000 kilobytes/3600 seconds
  PFS (Y/N): N
  Transform sets={
    TS-HR-EXEC,
  }
  Interfaces using crypto map CM-HR-EXEC:
    FastEthernet0/0
    FastEthernet2/0

EXEC#show access-lists ACL-HR-EXEC
Extended IP access list ACL-HR-EXEC
  10 permit ip 10.10.10.0 0.0.0.255 10.20.10.0 0.0.0.255
EXEC#show access-lists ACL-EXEC-HR
Extended IP access list ACL-EXEC-HR
  10 permit ip 10.20.10.0 0.0.0.255 10.10.10.0 0.0.0.255 (28 matches)
EXEC#show crypto isakmp policy

Global IKE policy
Protection suite of priority 10
  encryption algorithm: AES - Advanced Encryption Standard (256 bit keys).
  hash algorithm:      Secure Hash Standard
  authentication method: Pre-Shared Key
  Diffie-Hellman group: #5 (1536 bit)
  lifetime:            86400 seconds, no volume limit
Protection suite of priority 20
  encryption algorithm: AES - Advanced Encryption Standard (128 bit keys).
  hash algorithm:      Secure Hash Standard
  authentication method: Pre-Shared Key
  Diffie-Hellman group: #5 (1536 bit)
  lifetime:            28800 seconds, no volume limit
Protection suite of priority 30
  encryption algorithm: AES - Advanced Encryption Standard (128 bit keys).
  hash algorithm:      Secure Hash Standard
  authentication method: Pre-Shared Key
  Diffie-Hellman group: #2 (1024 bit)
  lifetime:            86400 seconds, no volume limit
Default protection suite
  encryption algorithm: DES - Data Encryption Standard (56 bit keys).
  hash algorithm:      Secure Hash Standard
--More--

```

```
HR#show crypto map interface FastEthernet0/0
Crypto Map "CM-HR-EXEC" 10 ipsec-isakmp
  Peer = 10.10.40.2
  Extended IP access list ACL-HR-EXEC
    access-list ACL-HR-EXEC permit ip 10.10.10.0 0.0.0.255 10.20.10.0 0.0.0.255
  Current peer: 10.10.40.2
  Security association lifetime: 4608000 kilobytes/3600 seconds
  PFS (Y/N): N
  Transform sets={
    TS-HR-EXEC,
  }
  Interfaces using crypto map CM-HR-EXEC:
    FastEthernet0/0

    FastEthernet2/0

HR#show access-lists ACL-HR-EXEC
Extended IP access list ACL-HR-EXEC
  10 permit ip 10.10.10.0 0.0.0.255 10.20.10.0 0.0.0.255 (29 matches)
HR#show access-lists ACL-EXEC-HR

HR#show crypto isakmp policy

Global IKE policy
Protection suite of priority 10
  encryption algorithm: AES - Advanced Encryption Standard (256 bit keys).
  hash algorithm:      Secure Hash Standard
  authentication method: Pre-Shared Key
  Diffie-Hellman group: #5 (1536 bit)
  lifetime:            86400 seconds, no volume limit
Default protection suite
  encryption algorithm: DES - Data Encryption Standard (56 bit keys).
  hash algorithm:      Secure Hash Standard
  authentication method: Rivest-Shamir-Adleman Signature
  Diffie-Hellman group: #1 (768 bit)
  lifetime:            86400 seconds, no volume limit
HR#
```

Requirement 02

```
MAN_HO#show ip vrf
Name                               Default RD      Interfaces
EXEC_VPN                          65001:2
HR_VPN                            65001:1
VPN-EXEC-RD                        65000:20       Fa2/0
VPN-HRL                           65000:10       Fa0/0.10
                                   Fa0/1
VPN-OTHER                          65000:30
VRF-EXEC-RND                      65000:300
VRF-OPS                           65000:100

MAN_HO#show ip vrf interfaces
Interface      IP-Address      VRF              Protocol
Fa2/0          10.20.40.10    VPN-EXEC-RD      up
Fa0/0.10       unassigned      VPN-HRL          down
Fa0/1          10.10.10.1     VPN-HRL          up

MAN_HO#show ip route vrf VPN-HRL

Routing Table: VPN-HRL
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/24 is subnetted, 4 subnets
C       10.10.10.0 is directly connected, FastEthernet0/1
B       10.10.20.0 [200/0] via 2.2.2.2, 01:48:26
B       10.10.30.0 [200/0] via 3.3.3.3, 01:48:41
B       10.40.10.0 [200/20] via 3.3.3.3, 01:48:41
MAN_HO#show mpls forwarding vrf VPN-HRL
Local  Outgoing  Prefix          Bytes tag  Outgoing   Next Hop
tag    tag or VC    or Tunnel Id    switched  interface
16     Aggregate  10.10.10.0/24[V]  0
MAN_HO#show bgp vpnv4 unicast all summary
BGP router identifier 1.1.1.1, local AS number 65000
BGP table version is 11, main routing table version 11
5 network entries using 700 bytes of memory
5 path entries using 340 bytes of memory
6/5 BGP path/bestpath attribute entries using 744 bytes of memory
5 BGP extended community entries using 200 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 2016 total bytes of memory
BGP activity 5/0 prefixes, 5/0 paths, scan interval 15 secs

Neighbor      V    AS MsgRcvd MsgSent  TblVer  InQ  OutQ Up/Down  State/PfxRcd
2.2.2.2       4 65000   113     115     11    0    0 01:49:03      1
3.3.3.3       4 65000   115     115     11    0    0 01:49:19      2
MAN_HO#
```

```
P#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
3.3.3.3	1	FULL/BDR	00:00:30	172.20.20.10	FastEthernet2/0
2.2.2.2	1	FULL/BDR	00:00:39	172.20.20.6	FastEthernet0/1
1.1.1.1	1	FULL/BDR	00:00:30	172.20.20.2	FastEthernet0/0

```
P#show mpls ldp neighbor
```

```
Peer LDP Ident: 3.3.3.3:0; Local LDP Ident 10.255.255.254:0
```

```
TCP connection: 3.3.3.3.646 - 10.255.255.254.47417
```

```
State: Oper; Msgs sent/rcvd: 203/204; Downstream
```

```
Up time: 02:49:26
```

```
LDP discovery sources:
```

```
FastEthernet2/0, Src IP addr: 172.20.20.10
```

```
Addresses bound to peer LDP Ident:
```

```
172.20.20.10 3.3.3.3
```

```
Peer LDP Ident: 2.2.2.2:0; Local LDP Ident 10.255.255.254:0
```

```
TCP connection: 2.2.2.2.646 - 10.255.255.254.38757
```

```
State: Oper; Msgs sent/rcvd: 204/204; Downstream
```

```
Up time: 02:49:26
```

```
LDP discovery sources:
```

```
FastEthernet0/1, Src IP addr: 172.20.20.6
```

```
Addresses bound to peer LDP Ident:
```

```
172.20.20.6 2.2.2.2
```

```
Peer LDP Ident: 1.1.1.1:0; Local LDP Ident 10.255.255.254:0
```

```
TCP connection: 1.1.1.1.646 - 10.255.255.254.29803
```

```
State: Oper; Msgs sent/rcvd: 138/137; Downstream
```

```
Up time: 01:51:09
```

```
LDP discovery sources:
```

```
FastEthernet0/0, Src IP addr: 172.20.20.2
```

```
Addresses bound to peer LDP Ident:
```

```
1.1.1.1 172.20.20.2
```

```
P#
```

```
P#ping 172.20.20.10
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 172.20.20.10, timeout is 2 seconds:
```

```
!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 12/19/28 ms
```

```
P#ping 172.20.20.6
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 172.20.20.6, timeout is 2 seconds:
```

```
!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/20/24 ms
```

```
P#ping 172.20.20.2
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 172.20.20.2, timeout is 2 seconds:
```

```
!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/20/24 ms
```

```
P#
```

```
HR#show ip interface brief | include FastEthernet2/0
FastEthernet2/0      10.10.11.2      YES manual up
HR#show ip ospf neighbor

HR#show ip route 10.20.10.0
Routing entry for 10.20.10.0/24
  Known via "static", distance 1, metric 0
  Routing Descriptor Blocks:
    * 10.10.40.2
      Route metric is 0, traffic share count is 1

HR#ping 10.20.10.2 source 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.10.2, timeout is 2 seconds:
Packet sent with a source address of 10.10.10.2
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/9/12 ms
HR#
```

```
EXEC#show ip interface brief | include FastEthernet2/0
FastEthernet2/0      10.20.11.2      YES manual up
EXEC#show ip ospf neighbor

EXEC#show ip route 10.10.10.0
Routing entry for 10.10.10.0/24
  Known via "static", distance 1, metric 0
  Routing Descriptor Blocks:
    * 10.10.40.1
      Route metric is 0, traffic share count is 1

EXEC#ping 10.10.10.2 source 10.20.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
Packet sent with a source address of 10.20.10.2
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/20/24 ms
EXEC#
```

Requirement 03

```
RND#show ip interface brief
Interface                IP-Address      OK? Method Status        Protocol
FastEthernet0/0          10.30.10.2      YES manual up             up
Serial0/0                 unassigned      YES NVRAM   administratively down down
FastEthernet0/1          10.30.11.2      YES manual up             up
Serial0/1                 unassigned      YES NVRAM   administratively down down
Serial0/2                 unassigned      YES NVRAM   administratively down down
Serial0/3                 unassigned      YES NVRAM   administratively down down
Serial1/0                 unassigned      YES NVRAM   administratively down down
Serial1/1                 unassigned      YES NVRAM   administratively down down
Serial1/2                 unassigned      YES NVRAM   administratively down down
Serial1/3                 unassigned      YES NVRAM   administratively down down
FastEthernet2/0           10.30.20.2      YES manual up             up
RND#
RND#ping 10.30.11.2 source FastEthernet0/1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.30.11.2, timeout is 2 seconds:
Packet sent with a source address of 10.30.11.2
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
RND#ping 10.30.10.2 source FastEthernet0/0

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.30.10.2, timeout is 2 seconds:
Packet sent with a source address of 10.30.10.2
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
RND#clear crypto sa
RND#
RND#ping 10.30.10.2 source FastEthernet0/0

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.30.10.2, timeout is 2 seconds:
Packet sent with a source address of 10.30.10.2
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
RND#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status

IPv6 Crypto ISAKMP SA

RND#show crypto ipsec sa

interface: FastEthernet0/0
  Crypto map tag: CM-EXEC-RND, local addr 10.30.10.2

  protected vrf: (none)
  local ident (addr/mask/prot/port): (10.20.20.0/255.255.255.0/0/0)
  remote ident (addr/mask/prot/port): (10.20.10.0/255.255.255.0/0/0)
  current_peer 10.20.10.2 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
    #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

    local crypto endpt.: 10.30.10.2, remote crypto endpt.: 10.20.10.2
    path mtu 1500, ip mtu 1500, ip mtu idb FastEthernet0/0
    current outbound spi: 0x0(0)
```

```

RND#clear crypto sa
RND#ping 10.30.11.2 source FastEthernet0/1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.30.11.2, timeout is 2 seconds:
Packet sent with a source address of 10.30.11.2
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
RND#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status

IPv6 Crypto ISAKMP SA

RND#show crypto ipsec sa

interface: FastEthernet0/0
  Crypto map tag: CM-EXEC-RND, local addr 10.30.10.2

  protected vrf: (none)
  local ident (addr/mask/prot/port): (10.20.20.0/255.255.255.0/0/0)
  remote ident (addr/mask/prot/port): (10.20.10.0/255.255.255.0/0/0)
  current_peer 10.20.10.2 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
    #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

    local crypto endpt.: 10.30.10.2, remote crypto endpt.: 10.20.10.2
    path mtu 1500, ip mtu 1500, ip mtu idb FastEthernet0/0
    current outbound spi: 0x0(0)

    inbound esp sas:

    inbound ah sas:

--More--

```



```
NCL#show ip interface brief | include FastEthernet2/0
FastEthernet2/0      10.30.20.1      YES manual up      up
NCL#
NCL#show ip route 10.20.10.0
Routing entry for 10.20.10.0/24
  Known via "static", distance 1, metric 0
  Routing Descriptor Blocks:
    * 10.10.50.1
      Route metric is 0, traffic share count is 1
NCL#ping 10.10.50.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.50.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/13/20 ms
NCL#
```

```
RND#show ip interface brief | include FastEthernet0/0|FastEthernet0/1|FastEthernet2/0
FastEthernet0/0      10.30.10.2      YES manual up      up
FastEthernet0/1      10.30.11.2      YES manual up      up
FastEthernet2/0      10.30.20.2      YES manual up      up
RND#show ip route 10.20.10.0
Routing entry for 10.20.10.0/24
  Known via "static", distance 1, metric 0
  Routing Descriptor Blocks:
    * 10.30.20.1
      Route metric is 0, traffic share count is 1
RND#ping 10.30.20.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.30.20.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
RND#
```

Requirement 04

```
HR#show access-lists ACL-SQL-HR
Extended IP access list ACL-SQL-HR
 10 permit tcp 10.10.10.0 0.0.0.255 host 10.20.30.10 eq 1433
 15 permit icmp 10.10.10.0 0.0.0.255 host 10.20.30.10
 20 deny ip any host 10.20.30.10
 30 permit ip any any
HR#show run interface FastEthernet0/1
Building configuration...

Current configuration : 145 bytes
!
interface FastEthernet0/1
 description HR LAN
 ip address 10.10.10.2 255.255.255.0
 ip access-group ACL-SQL-HR in
 speed 100
 full-duplex
end

HR#show ip route 10.20.30.0
Routing entry for 10.20.30.0/24
  Known via "static", distance 1, metric 0
  Routing Descriptor Blocks:
 * 10.10.11.1
   Route metric is 0, traffic share count is 1

HR#ping 10.20.30.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.30.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 20/20/24 ms
HR#
```

```
LDS#show ip interface brief | include FastEthernet0/1
FastEthernet0/1      10.20.30.1      YES manual up
LDS#show run interface FastEthernet0/1
Building configuration...

Current configuration : 178 bytes
!
interface FastEthernet0/1
 description To PROD1 CE (VPN-HRL)
 ip address 10.20.30.10 255.255.255.0 secondary
 ip address 10.20.30.1 255.255.255.0
 duplex auto
 speed auto
end
```

```
EXEC#show ip interface brief | include FastEthernet0/1
FastEthernet0/1      10.20.10.254    YES manual up
EXEC#ping 10.20.30.10 source FastEthernet0/1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.30.10, timeout is 2 seconds:
Packet sent with a source address of 10.20.10.254
.....
Success rate is 0 percent (0/5)
EXEC#
```

Requirement 05

```
NCL#show run interface FastEthernet0/1
Building configuration...

Current configuration : 199 bytes
!
interface FastEthernet0/1
 ip vrf forwarding PRODUCTION2
 ip address 192.168.70.1 255.255.255.0
 ip access-group BLOCK_PRODUCTION2_TO_EXEC in
 duplex auto
 speed auto
 crypto map RND-EXEC-MAP
end

NCL#show access-lists BLOCK_PRODUCTION2_TO_EXEC
Extended IP access list BLOCK_PRODUCTION2_TO_EXEC
 10 deny ip 192.168.70.0 0.0.0.255 192.168.40.0 0.0.0.255
 20 permit ip any any
NCL#
NCL#
```

```

NCL#show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
10.255.255.254  1     FULL/DR         00:00:38    172.20.20.9    FastEthernet0/0
%OSPF: Router process 300 is not running, please configure a router-id
10.10.30.254   1     FULL/DR         00:00:37    10.10.30.2     FastEthernet0/1
%OSPF: Router process 330 is not running, please configure a router-id
NCL#
*Mar  1 00:00:32.363: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1, changed state to down
NCL#
*Mar  1 00:00:35.687: %LDP-5-NBRCHG: LDP Neighbor 10.255.255.254:0 (1) is UP
NCL#
*Mar  1 00:00:37.067: %BGP-5-ADJCHANGE: neighbor 1.1.1.1 Up
NCL#
*Mar  1 00:00:39.655: %BGP-5-ADJCHANGE: neighbor 2.2.2.2 Up
NCL#show access-lists ACL-BLOCK-PROD2-EXEC
Extended IP access list ACL-BLOCK-PROD2-EXEC
 10 deny ip 10.40.20.0 0.0.0.255 10.20.10.0 0.0.0.255
 20 permit ip any any
NCL#show run interface FastEthernet2/1
Building configuration...

Current configuration : 147 bytes
!
interface FastEthernet2/1
 description Link to PRODUCTION-2 CE (f0/0)
 ip vrf forwarding VPN-PROD2
 ip access-group ACL-BLOCK-PROD2-EXEC in
end

NCL#ping vrf VPN-PROD2 10.20.10.5 source 10.40.20.5

% VRF does not have a usable source address
NCL#ping vrf VPN-PROD2 8.8.8.8 source 10.40.20.5

% VRF does not have a usable source address
NCL#

```

Requirement 06

```

LDS#show ip dhcp pool

Pool PROD1_POOL :
Utilization mark (high/low)      : 100 / 0
Subnet size (first/next)         : 0 / 0
Total addresses                   : 254
Leased addresses                  : 0
Pending event                     : none
1 subnet is currently in the pool :
Current index      IP address range      Leased addresses
10.3.3.1           10.3.3.1 - 10.3.3.254                0

Pool PROD1-LAN :
Utilization mark (high/low)      : 100 / 0
Subnet size (first/next)         : 0 / 0
Total addresses                   : 254
Leased addresses                  : 0
Pending event                     : none
1 subnet is currently in the pool :
Current index      IP address range      Leased addresses
10.20.30.1         10.20.30.1 - 10.20.30.254                0

LDS#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/      Lease expiration        Type
                   Hardware address/
                   User name
LDS#

```

```

HR#ping 10.20.10.2 source 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.10.2, timeout is 2 seconds:
Packet sent with a source address of 10.10.10.2
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/26/44 ms
HR#
HR#ping 10.10.40.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.40.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/19/20 ms
HR#show ip dhcp pool

Pool HR-POOL :
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)         : 0 / 0
  Total addresses                   : 126
  Leased addresses                  : 0
  Pending event                     : none
  1 subnet is currently in the pool :
    Current index   IP address range      Leased addresses
    10.0.20.1       10.0.20.1 - 10.0.20.126 0

Pool HR-LAN :
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)         : 0 / 0
  Total addresses                   : 254
  Leased addresses                  : 0
  Pending event                     : none
  1 subnet is currently in the pool :
    Current index   IP address range      Leased addresses
    10.10.10.1      10.10.10.1 - 10.10.10.254 0
HR#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration      Type
                  Hardware address/
                  User name
HR#

```

```

EXEC#show ip dhcp pool

Pool EXEC-POOL :
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)         : 0 / 0
  Total addresses                   : 62
  Leased addresses                  : 0
  Pending event                     : none
  1 subnet is currently in the pool :
    Current index   IP address range      Leased addresses
    10.0.10.1       10.0.10.1 - 10.0.10.62 0

Pool EXEC-LAN :
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)         : 0 / 0
  Total addresses                   : 254
  Leased addresses                  : 0
  Pending event                     : none
  1 subnet is currently in the pool :
    Current index   IP address range      Leased addresses
    10.20.10.1      10.20.10.1 - 10.20.10.254 0
EXEC#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration      Type
                  Hardware address/
                  User name
EXEC#
EXEC#ping 10.10.40.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.40.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/24/40 ms
EXEC#

```

```
NCL#show ip dhcp pool
```

```
Pool LOG_POOL :
```

```
Utilization mark (high/low) : 100 / 0
```

```
Subnet size (first/next) : 0 / 0
```

```
Total addresses : 254
```

```
Leased addresses : 0
```

```
Pending event : none
```

```
1 subnet is currently in the pool :
```

Current index	IP address range	Leased addresses
10.4.4.1	10.4.4.1 - 10.4.4.254	0

```
Pool RND_A_POOL :
```

```
Utilization mark (high/low) : 100 / 0
```

```
Subnet size (first/next) : 0 / 0
```

```
Total addresses : 254
```

```
Leased addresses : 0
```

```
Pending event : none
```

```
1 subnet is currently in the pool :
```

Current index	IP address range	Leased addresses
10.5.5.1	10.5.5.1 - 10.5.5.254	0

```
Pool RND_B_POOL :
```

```
Utilization mark (high/low) : 100 / 0
```

```
NCL#show ip dhcp binding
```

```
Bindings from all pools not associated with VRF:
```

IP address	Client-ID/ Hardware address/ User name	Lease expiration	Type
------------	--	------------------	------

```
NCL#
```